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Placement Director: Oleg Itskhoki — itskhoki@fas.harvard.edu — 617-495-2144
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Education

Harvard University
Ph.D. Economics, 2021 to 2027 (expected)

University of Chicago
B.A. in Economics (with Honors), 2019
B.S. in Mathematics (with Specialization in Economics), 2019
B.S. in Statistics, 2019

Fields

Industrial Organization, Innovation, Digitization & AI

References

Shane Greenstein (Chair)
sgreenstein@hbs.edu

Robin Lee
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Ariel Pakes
apakes@fas.harvard.edu

Jesse Shapiro
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Fellowships & Awards

Dissertation Completion Fellowship, Harvard University	2026
Humane Studies Fellowship, Institute for Humane Studies	2026
John and Kiendl Gordon Funding Award, Harvard University	2026
Lab for Economic Applications and Policy Funding Award, Harvard University	2025-2026
Becker-Friedman Institute Academic Award, University of Chicago	2019
David S. Hu Thesis Award, University of Chicago	2019
Phi Beta Kappa (Junior Year), University of Chicago	2018
Student Marshal, University of Chicago	2018
Dougan Scholar Certificate Program, Chicago Booth School of Business	2016-2019

Teaching

Teaching Fellow for Professor Jesse Shapiro, *Computing for Economists* (Graduate)	2024-2025
Teaching Fellow for Professor Robin Lee, *Industrial Organization* (Undergraduate)	2024-2025

Job Market Paper

Returns to Science of Centrally Provisioned GPUs and Optimal Allocation (with Connie Xu)

Funders of research must consider where the marginal dollar is most productive and where it is least likely to displace private investment. We study the allocation of federal computing grants to U.S. universities by estimating the returns to science of university-level computing infrastructure and modeling the crowd-out of private investment. We construct an original dataset measuring Central Processing Unit (CPU) and Graphics Processing Unit (GPU) capacity as well as research publications at U.S. R1 universities over time. Exploiting the large and lumpy nature of GPU capacity increases to identify its effects, we find that AI and deep-learning publications increase more at less-resourced universities. We then estimate a structural production function where multiple scientific fields share a central computing cluster, and derive universities' investment responses to external grants. Compared to a publication-maximizing benchmark, the observed portfolio of National Science Foundation (NSF) cyberinfrastructure grants captures 39% of the publication gain attainable with the same budget. The publication-maximizing allocation shifts dollars from universities with medium-sized endowments toward both tails of the endowment distribution. However, the NSF's observed allocation is more geographically dispersed than the publication-maximizing allocation, highlighting tradeoffs in balancing efficiency versus equity.

Publications

Greenstein, N., Fontana, R., Greenstein, S., Zhang, R., Zhang, H., Olivarez, O., and Kim, D.Y. Wireless communications equipment markets: evolution, classification, and measurement. *Telecommunications Policy* 50(6), 103209 (2026). https://doi.org/10.1016/j.telpol.2026.103209

Zhang, R. "Building Up Research Capacity at Minority Institutions." Commissioned paper for *A Plan to Promote Defense Research at Minority-Serving Institutions* (2024).

Chetty, R., Jackson, M.O., Kuchler, T. *et al.* Social capital I: measurement and associations with economic mobility. *Nature* **608**, 108–121 (2022). https://doi.org/10.1038/s41586-022-04996-4

Chetty, R., Jackson, M.O., Kuchler, T. *et al.* Social capital II: determinants of economic connectedness. *Nature* **608**, 122–134 (2022). https://doi.org/10.1038/s41586-022-04997-3

Working Papers

Secrecy, Spectrum, and Certification: Evidence from Wireless Electronic Devices
(with Roberto Fontana, Shane Greenstein, and Do Yoon Kim)

This paper examines the strategic decisions of firms to withhold information in innovation-intensive markets, highlighting that secrecy contains distinct temporal dimensions. Using all FCC equipment filings from 2001 to 2021, we analyze firms' choices between long-term confidentiality (LTC), which permanently conceals technical information such as block diagrams, and short-term confidentiality (STC), which temporarily delays the public release of marketing information to ensure against launch and coordination risks in the commercialization process. LTC use reflects appropriability concerns and institutional complexities—technically complex products and competitive markets are 12 to 35% more likely to adopt it. By contrast, STC is driven by the commercialization timing of branded products by large firms. Large, domestic, patent-owning firms with regulatorily complex products are more likely to use STC. Technical aspects of the product do not predict STC usage. An exposure-based event study further shows that firms with a high ex-ante propensity to use STC bring products to market faster after policy enactment, with no changes in product complexity or volume.

Managing the Rising Cost of Science: Supplier Consolidation in the Academic Life Sciences
(with Connie Xu)

Scientists cannot easily substitute away from the inputs their experiments depend on or pass on higher costs, leaving them vulnerable to rising prices in increasingly concentrated markets for life sciences inputs. How much science is lost when these inputs become more expensive, and which scientists are better able to adjust? We use the 2014 merger of two large life sciences suppliers as a shock to the prices principal investigators (PIs) pay for these inputs, drawing on novel purchase-level procurement records from U.S. public universities linked to PIs' publications. Prices in affected markets rose by almost 20 percent, while quantities declined only slightly and temporarily, consistent with highly inelastic demand. PIs facing larger increases in their input costs published less. For the average PI, input costs rose by about 2.8 percent and research output fell by about 1.2 percent, implying an elasticity of −0.42. Higher input prices therefore reduce scientific production, not just raise research costs. The output losses fell disproportionately on early-career PIs. Financial resources do not fully explain this difference. Instead, our findings suggest that experience helps PIs adjust to higher prices through tacit knowledge accumulated during training and years of running a lab.

Seminars & Conferences

American Economic Association Annual Meeting (AEA ASSA)	2027
Roundtable for Engineering Entrepreneurship Research Conference (REER)	2026
NBER Productivity Seminar	
Disclosure, Information Sharing, and Secrecy Workshop (DISS)	
Research Conference on Communications, Information and Internet Policy (TPRC)	
European Association for Research in Industrial Economics Conference (EARIE)	
Academy of Management Annual Meeting (AOM)	
Danish Research Unit for Industrial Dynamics Conference (DRUID)	
New England Economics PhD Conference	
International Industrial Organization Conference (IIOC)	
Econometric Society Summer School in Dynamic Structural Econometrics (attendee)	2025

	Harvard Seminar in the Economics of Science & Engineering	2024
	Machine Learning in Economics Summer Institute (attendee)	
	NBER Digital Economics and AI Tutorial (attendee)	2023
	NBER Innovation Research Bootcamp (attendee)	
Employment	Research Assistant to Shane Greenstein, Harvard Business School	2022-2025
	Research Assistant to Raj Chetty and Nathaniel Hendren, Harvard University	2019-2021
	Research Assistant to James Heckman, University of Chicago	2016-2018
	Summer Analyst in Mergers and Acquisitions Advisory, Lazard	2018
Academic Service	Organizer, Harvard Industrial Organization Reading Group	2024-2026
	Mentor, MIT & Harvard Economics Application Assistance and Mentoring Program	2022-2023
Languages	English (native), Mandarin Chinese (fluent), French (basic)	
Citizenship	Canada	